

Designing the Boundary

Constitutive Markov-Blanket Inversion in Individual and Collective Drones

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Abstract

Active-inference models usually assume the Markov blanket through which an agent senses and acts. This paper asks a stronger design-theoretic question: can an artifact infer all four blanket roles and then select interventions that make a different boundary empirically true? The physical causal organization, $\mathcal{C}_t$, is separated from the artifact's posterior belief about that organization, $q_t(M)$. Each drone maintains an exact joint posterior over all $8!/(2!)^4 = 2520$ assignments of eight modules to internal, sensory, active, and external roles; learns transition and intervention models online; and selects control and design actions as a joint policy by expected free energy (EFE). Shielding removes unscreened interior–exterior drift channels, while routing exchanges the physical modules occupying causal ports. The intervention is therefore constitutive rather than merely diagnostic.

The paper has two experimental parts. Part I tests an individual drone on the three verbs central to Possati's *Design for Entropy*: construct, stabilize, and transform. Across 360 frozen missions, the constitutive agent succeeds in $\sim 80\%$, $\sim 90\%$, and $\sim 100\%$ of the respective tasks, versus $\sim 0\%$, $\sim 0\%$, and $\sim 0\%$ for a diagnostic-only control. Part II asks whether two drones can infer and enact a group-level blanket. No unilateral proposal can change the group organization; evidence pooling and a bilateral EFE handshake are required. Across 420 additional missions, the collective agent succeeds in 80% of assembly trials, 90% of repair trials, and 100% of reconfiguration trials. An equal-bandwidth communication control succeeds in 0% of all three. These results establish a bounded proof of concept for causal boundary design and operational collective individuation. They do not establish open-ended self-individuation, hardware autonomy, or the ontological identity of a drone swarm.

Keywords: active inference; Markov blanket; expected free energy; design theory; structural learning; path space; causal intervention; collective agency; multi-agent systems.

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1 Introduction

Autonomous systems normally learn inside an interface that engineers have already fixed. Sensors are inputs, motors are outputs, software variables are internal, and everything else is environment. Parameters may change, but the partition that makes those parameters meaningful is treated as invariant. This assumption is fragile whenever channels fail, resources are rerouted, or the task requires a different coupling between artifact and world.

The free-energy principle (FEP) and active inference provide a statistical vocabulary for questioning that assumption. An adaptive system can be described by exterior states Y , interior states X , and boundary states $B = (S, A)$, where sensory states S mediate exterior influence on the interior and active states A mediate interior influence on the exterior [9, 10, 16, 13, 27]. At an instant, a Markov blanket satisfies

$$Y_t \perp\!\!\!\perp X_t \mid B_t. \quad (1.1)$$

The stronger path-space statement is

$$Y_{[0,T]} \perp\!\!\!\perp X_{[0,T]} \mid B_{[0,T]}, \quad (1.2)$$

which rules out hidden temporal channels that an instantaneous graph can miss [14, 29].

Beck and Ramstead treat blanket identification as generative-model inversion: the system infers which microscopic variables occupy macroscopic functional roles from their dynamics [2]. Possati’s proposal in *Design for Entropy* reverses the practical direction of that inversion [26]. If model inversion can disclose a statistical boundary, design can select interventions that construct, stabilize, or transform the couplings that make a boundary true. The reversal is not a change of label. It changes the object of action from a belief about organization to the organization itself.

This distinction yields two nested research questions:

RQ1—individual constitution. Can a computational artifact jointly infer all four roles of its Markov-blanket organization, learn how structural interventions work, and use EFE to select control and design actions that *cause* a path-space boundary to be constructed, stabilized, or transformed—with measurable consequences for viability?

RQ2—collective constitution. Can two partially informed artifacts pool structural evidence and jointly enact a group-level boundary that no member can constitute alone, and does that boundary improve their strategic performance relative to individual, fixed-team, random, and equal-bandwidth communication controls?

The question is deliberately discriminating. A model does not count as boundary design merely because its estimate $q_t(M)$ changes. It counts only if a chosen design action d_t changes the physical causal organization:

$$\frac{\partial \mathcal{C}_{t+1}}{\partial d_t} \neq 0, \quad (1.3)$$

or, for discrete actions, if two admissible actions under the same exogenous realization produce different $\mathcal{C}_{t+1}$. This interventionist criterion follows the ordinary causal distinction between observing a variable and changing the mechanism that generates it [23, 24].

The contribution is fourfold. First, the model separates epistemic boundary beliefs from the physical causal organization on which those beliefs bear. Second, it makes discovery complete: internal, sensory, active, and external roles are inferred jointly as a constrained partition, rather than treating one role as an unexamined anchor. Third, it learns intervention efficacy and selects motor and structural actions in one EFE policy. Fourth, it extends boundary design across scale: a pair of drones must establish a shared blanket through evidence pooling and bilateral agreement.

The paper proceeds from the common conceptual framework to two experimental parts. Sections 2–3 introduce the FEP, active inference, and Possati’s design-theoretic inversion. Part I specifies and

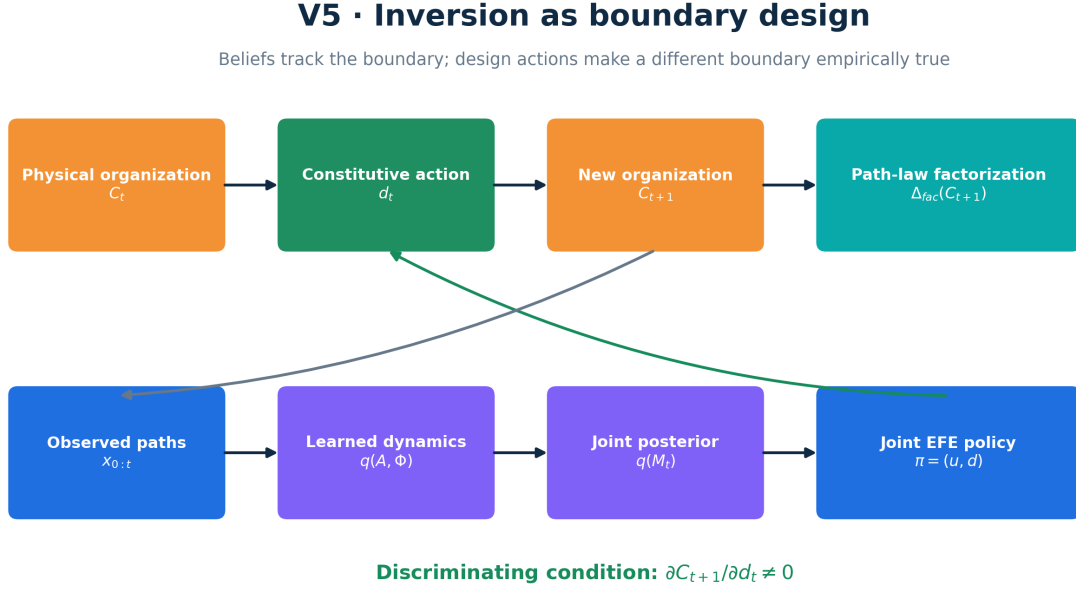


Figure 1: V5 separates physical causal organization C_t from the posterior $q_t(M)$. The lower loop learns and evaluates; the upper loop is changed by constitutive design action.

tests individual boundary design. Part II defines a genuinely joint boundary and tests its strategic value. The discussion separates computational evidence for operational individuation from stronger metaphysical claims. Appendices provide the path-space derivation, algorithms, parameter register, complete result tables, and reproducibility information.

2 FEP, Active Inference, and Design

2.1 Variational free energy

Let o denote observations, z hidden states, $p(o, z)$ a generative model, and $q(z)$ a variational density. Variational free energy is

$$F[q; o] = \mathbb{E}_{q(z)}[\ln q(z) - \ln p(o, z)] \quad (2.1)$$

$$= D_{\text{KL}}[q(z) \parallel p(z \mid o)] - \ln p(o). \quad (2.2)$$

Because the KL divergence is nonnegative, $F \geq -\ln p(o)$. Minimizing F with respect to q makes the approximate posterior approach the Bayesian posterior. This identity is a statement about inference; stronger claims about organisms or artifacts require additional assumptions about dynamics, ergodicity, and the partition of states [10, 13, 18, 21, 4, 5].

Under active inference, actions are included in policies π . A standard expected-free-energy functional for future outcomes o_τ and states z_τ can be written

$$G_\tau(\pi) = \mathbb{E}_{q(o_\tau, z_\tau | \pi)}[\ln q(z_\tau | \pi) - \ln p(o_\tau, z_\tau)] \quad (2.3)$$

$$\approx \underbrace{\mathbb{E}_{q(o_\tau | \pi)}[-\ln p^*(o_\tau)]}_{\text{risk}} + \underbrace{\mathbb{E}_{q(z_\tau | \pi)}[\text{H}(p(o_\tau \mid z_\tau))]}_{\text{ambiguity}} - \underbrace{\text{I}_q(z_\tau; o_\tau \mid \pi)}_{\text{epistemic value}}, \quad (2.4)$$

where $p^*(o)$ encodes preferred outcomes. Equation (2.4) matters because policy selection is not only reward maximization. Policies can be valuable because they are expected to resolve uncertainty [20, 8, 21, 11, 12, 25, 30].

2.2 Markov blankets and paths

A blanket is sometimes presented as a static graphical separator. That is insufficient here. The drone acts and learns through histories, and hidden memory can preserve interior–exterior dependence even when Equation (1.1) holds at a single time. Sakthivadivel’s path-space analysis treats the boundary as a factorization of regular conditional laws [29]:

$$P(dY_{[0,T]}, dX_{[0,T]} \mid B_{[0,T]}) = P(dY_{[0,T]} \mid B_{[0,T]})P(dX_{[0,T]} \mid B_{[0,T]}). \quad (2.5)$$

When exact factorization fails, the residual dependence is the conditional mutual information

$$I(Y_{[0,T]}; X_{[0,T]} \mid B_{[0,T]}) = \mathbb{E}_B D_{\text{KL}} \left[P_{Y,X|B} \parallel P_{Y|B} P_{X|B} \right]. \quad (2.6)$$

For diffusions, Girsanov’s theorem connects changes of path law to quadratic control energy. This supplies the mathematical reason V5 measures direct unscreened drift rather than only comparing role labels.

2.3 From inference to design theory

Existing work has already connected active inference to niche construction, material engagement, cultural affordances, human–computer interaction, and the design of intelligible artificial systems [3, 6, 7, 31, 1, 15, 19]. These approaches show how an agent changes its environment, selects observations, or externalizes generative models. They are essential antecedents, but most begin with an agent–environment partition that remains analytically fixed.

Multi-agent active-inference research has modeled generalized synchrony, belief alignment, collective intelligence, and social niche construction [22, 17, 28]. Those models motivate the second experiment but do not by themselves settle its question. Synchrony or shared prediction can arise between agents whose individual boundaries remain fixed. Part II instead makes the group partition a latent and controllable variable and requires that a group-level intervention be irreducibly bilateral.

Possati’s difference is the level at which design operates. The primary design variable is the partition itself: which states mediate inward and outward influence, which dependencies are screened off, and which histories become conditionally independent. This is not simply “active inference applied to design.” It is a design-theoretic interpretation of individuation in which an interface is an achieved causal-statistical organization.

Three operations from the book receive explicit computational counterparts:

- **Precision crafting** becomes the selection, gating, routing, and calibration of sensor and actuator modules. Precision is not only an inference weight; it is changed by altering which physical module occupies a boundary port.
- **Curiosity sculpting** becomes the EFE-guided choice of structural perturbations. The drone selects which coupling to test or change because of its expected epistemic and pragmatic value.
- **Prediction embedding** becomes priors over complete role partitions together with learned transition and intervention likelihoods. Expectations are embedded in a material routing architecture, not held only as symbolic forecasts.

3 Inversion as the Design of a Boundary

3.1 Detection, tracking, and constitution

Let M_t be a hypothesis about roles and let $\mathcal{C}_t$ be the actual causal organization. Detection asks whether $q_t(M)$ identifies a fixed $\mathcal{C}_t$. Tracking asks whether $q_t(M)$ follows exogenous changes in $\mathcal{C}_t$.

Three claims that must not be conflated

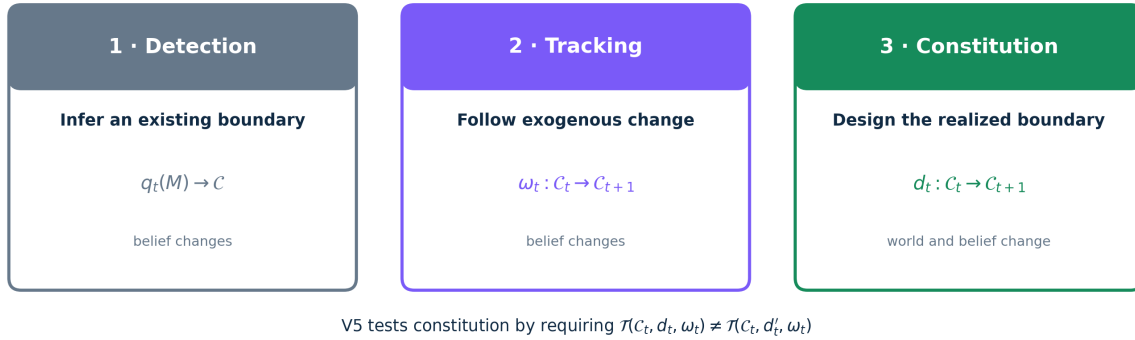


Figure 2: Three increasingly strong meanings of structural inversion. Detection estimates a fixed organization; tracking follows an organization changed from outside; constitution closes the loop by allowing the selected design action to alter the organization that generates future paths.

Constitution asks whether the artifact selects d_t such that

$$C_{t+1} = \mathcal{T}(C_t, d_t, \omega_t), \quad (3.1)$$

where ω_t collects exogenous disturbances, and such that the selected action changes the causal relations that define the boundary.

This separation corrects the central limitation of V4. In that model, a script exchanged thermal and optic-flow roles and probes returned noisy role cues from a known likelihood. The agent could detect and track the script, but the intervention did not create or remodel the boundary. V5 contains no exogenous role exchange. Exogenous events can damage precision or open a causal breach, but only a drone-selected routing or shielding action can produce the repaired or transformed organization.

3.2 The narrow thesis tested here

The model tests a bounded proposition:

Given a finite candidate state space, a finite repertoire of physically effective interventions, and online evidence about dynamics and precision, a joint EFE policy can select actions that improve path-law factorization and viability more reliably than an otherwise identical diagnostic policy.

This proposition is weaker than open-ended self-construction. It is also stronger than structural classification, because the dependent variable is the causal process itself.

3.3 Why this is specifically Possati's inversion

Beck and Ramstead invert a generative model in order to identify a macroscopic organization: observed microscopic dynamics provide evidence for which variables occupy blanket roles [2]. Possati retains that inferential movement but adds its practical reverse. Once a provisional organization has been inferred, an artifact can act on the couplings that support it. The output of model inversion becomes an input to design; design then changes the data-generating organization and initiates another round of inversion. The resulting loop is

$$\text{paths} \longrightarrow q_t(M) \longrightarrow d_t \longrightarrow C_{t+1} \longrightarrow \text{new paths}. \quad (3.2)$$

This is the book’s discriminating claim in computational form. The boundary is neither merely given by the engineer nor merely discovered by the agent. It is a revisable achievement whose causal support can be constructed, maintained, or redirected.

The three design notions used throughout the paper are correspondences, not decorative analogies. *Precision crafting* is implemented by sensor and actuator selection, gating, routing, and calibration. *Curiosity sculpting* is implemented by EFE-selected structural perturbations whose epistemic value depends on the current partition posterior. *Prediction embedding* is implemented by priors over complete role assignments and by learned likelihoods over dynamics and intervention outcomes. In Part II these operations are distributed across two bodies: precision is jointly routed, curiosity must be agreed, and predictions are embedded in a pooled group model.

Part I

Individual Boundary Design

4 Complete Mathematical Model

4.1 State space and four-role partitions

There are $n = 8$ generic modules $V = \{1, \dots, 8\}$. A role map is

$$r : V \rightarrow \{I, S, A, E\}. \quad (4.1)$$

The experiment fixes two modules per role. The hypothesis space is therefore

$$\mathcal{M} = \left\{ r : \sum_{j=1}^8 \mathbf{1}[r(j) = k] = 2, \quad k \in \{I, S, A, E\} \right\}, \quad |\mathcal{M}| = \frac{8!}{(2!)^4} = 2520. \quad (4.2)$$

The agent maintains a joint posterior $q_t(M)$ over $\mathcal{M}$. Marginal role probabilities are derived, not independently normalized:

$$q_t(r_j = k) = \sum_{M \in \mathcal{M}} q_t(M) \mathbf{1}[M(j) = k]. \quad (4.3)$$

Consequently, every posterior sample is a complete valid partition. The two nodes within each role are exchangeable; evaluation is therefore insensitive to their order.

4.2 Physical causal organization

The causal state is

$$\mathcal{C}_t = (r_t, W_t, L_t, \rho_t), \quad (4.4)$$

where r_t is the physical port assignment, $W_t \in \mathbb{R}^{8 \times 8}$ is a directed drift matrix, L_t is a set of unscreened links, and $\rho_t \in [0, 1]^8$ is module precision. Columns of W are causes and rows are effects. A clean template contains the directed cycle

$$E \longrightarrow S \longrightarrow I \longrightarrow A \longrightarrow E, \quad (4.5)$$

plus stable within-role recurrence. All other cross-role entries are zero. The template is scaled so its spectral radius is at most 0.82.

The module dynamics are linear Gaussian:

$$x_{t+1} = W(\mathcal{C}_t)x_t + b_A(r_t)u_t + b_E(r_t)w_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_t). \quad (4.6)$$

Here u_t is drone control, w_t is an autocorrelated wind disturbance, b_A injects control into modules currently occupying active ports, and b_E injects wind into modules occupying external ports.

4.3 A computable path-space gap

Let W_{IE} and W_{EI} be the direct drift blocks between current internal and external states. In a linear diffusion with isotropic innovation scale σ , removing those blocks changes the conditional path law. The quadratic Girsanov/Foellmer energy yields the benchmark measure

$$\Delta_{\text{fac}}(\mathcal{C}_t) = \frac{1}{2\sigma^2} \left(\|W_{IE}\|_F^2 + \|W_{EI}\|_F^2 \right). \quad (4.7)$$

It is zero for the simulated clean blanket and positive for a direct interior–exterior drift channel. In this controlled linear-Gaussian setting, Equation (4.7) is a local relative-entropy-rate surrogate for the path conditional mutual information in Equation (2.6). It is not claimed to be a universal metric for nonlinear physical systems.

4.4 Constitutive interventions

The design action is $d_t \in \mathcal{D} = \{d^0, d_{ij}^{\text{shield}}, d_{ij}^{\text{swap}}\}$. The actions have persistent effects:

$$d_{ij}^{\text{shield}} : (W_{ij}, W_{ji}) \mapsto (0, 0) \quad \text{when } (i, j) \in L_t, \quad (4.8)$$

$$d_{ij}^{\text{swap}} : (r_t(i), r_t(j)) \mapsto (r_t(j), r_t(i)). \quad (4.9)$$

Swapping changes which physical modules occupy the relevant causal ports while preserving exactly two nodes per role. Shielding can make Equation (2.5) true by eliminating a bypass channel.

The diagnostic-only control evaluates the same nominal actions and pays the same decision costs, but its transition is

$$\mathcal{T}_{\text{diag}}(\mathcal{C}_t, d_t, \omega_t) = \mathcal{T}_{\text{diag}}(\mathcal{C}_t, d^0, \omega_t). \quad (4.10)$$

This is the crucial negative control.

4.5 Learned dynamics and structural posterior

From a rolling window of transitions, the agent estimates W by ridge regression. With design matrix $Z = [X, U, \mathbf{1}]$ and successor matrix X^+ ,

$$\hat{\Theta}_t = (Z^\top Z + \lambda I)^{-1} Z^\top X^+, \quad \hat{W}_t = (\hat{\Theta}_{1:8})^\top. \quad (4.11)$$

For every $M \in \mathcal{M}$, let T_M denote its directed role template. The structural log score is

$$\ell_t(M) = -\tau \|\hat{W}_t\| - T_M \|_F^2 + \eta_A s_A(M; \hat{\Theta}_t) + \eta_E s_E(M; \hat{\Sigma}_t), \quad (4.12)$$

where s_A measures alignment between learned control response and active roles, and s_E measures alignment between learned innovation variance and external roles. Thus semantic orientation comes from learned response and noise statistics, not a supplied table of role cues. The posterior is

$$q_t(M) = \frac{\exp \ell_t(M)}{\sum_{M' \in \mathcal{M}} \exp \ell_t(M')}. \quad (4.13)$$

A small hazard mixture prevents premature certainty when the organization changes.

4.6 Learned intervention model

For each intervention family $k \in \{\text{shield}, \text{swap}\}$, let θ_k be the probability that an attempted action produces an observable causal change. V5 uses

$$\theta_k \sim \text{Beta}(a_k, b_k), \quad (a_k, b_k) \leftarrow (a_k + c_t, b_k + 1 - c_t), \quad (4.14)$$

where $c_t = 1$ exactly when the pre/post causal matrices differ. The predictive success probability and variance are

$$\hat{p}_k = \frac{a_k}{a_k + b_k}, \quad \text{Var}(\theta_k) = \frac{a_k b_k}{(a_k + b_k)^2 (a_k + b_k + 1)}. \quad (4.15)$$

The prior is Beta(1, 1). This model is intentionally modest: it learns family-level causal efficacy rather than receiving a known role-conditioned probe likelihood.

4.7 Precision and viability

Let

$$\bar{\rho}_S(t) = \frac{1}{2} \sum_{j:r_t(j)=S} \rho_j(t), \quad \bar{\rho}_A(t) = \frac{1}{2} \sum_{j:r_t(j)=A} \rho_j(t). \quad (4.16)$$

The cross-track state y_t evolves as

$$y_{t+1} = 0.91y_t + \delta(\tilde{w}_t + 1.22\bar{\rho}_A(t)u_t) + \xi_t, \quad \tilde{w}_t \sim \mathcal{N}\left(w_t, \frac{0.20^2}{\bar{\rho}_S(t)^2}\right). \quad (4.17)$$

Poor sensory routing therefore increases estimation noise; poor active routing attenuates control. A collision occurs at $|y_t| > 2.35$.

4.8 Joint expected-free-energy policy

A policy is the pair $\pi = (u, d)$. V5 scores all admissible pairs together:

$$G_t(u, d) = R_t(u) + \alpha \mathcal{A}_t(u) + c(d) \quad (4.18)$$

$$- \hat{p}_{k(d)} \left[\beta_s V_t^{\text{struct}}(d) + \beta_p V_t^{\text{precision}}(d) \right] - \beta_I \text{IG}_t(d). \quad (4.19)$$

R_t is predicted cross-track and control risk; $\mathcal{A}_t$ is predictive ambiguity from transition residuals; $c(d)$ is intervention cost; V_t^{struct} is the expected removal of an I – E coupling under the role posterior; $V_t^{\text{precision}}$ is the expected benefit of routing a precise module into a boundary port; and IG combines local role entropy with intervention-model uncertainty. Policies are sampled from

$$q(\pi) \propto \exp[-G_t(\pi)/\gamma]. \quad (4.20)$$

Equation (4.19) is “genuine EFE” in the operational sense that pragmatic risk, ambiguity, and expected information gain all alter policy probability. It remains a finite-horizon approximation, not an exact solution of a continuous-time variational control problem.

5 Experimental Design

5.1 Three discriminating tasks

Construction. Each mission starts with three direct I – E links. The agent constructs a boundary only by selecting the corresponding shielding actions. Successful construction requires a final gap below 10% of its initial value and no collision.

Stabilization. The mission begins with a clean boundary. At a seed-random time between steps 27 and 38, one direct I – E channel opens. Roles do not change. Successful stabilization requires a final gap below 0.25 and no collision.

Transformation. The mission begins with a clean boundary. At a seed-random time between steps 31 and 42, one current boundary module loses precision and one external module becomes

Condition	Definition
Constitutive	Joint EFE control and physically effective shield/swap actions.
Diagnostic only	Same nominal choices, but design actions cannot change $\mathcal{C}_t$.
Factorized	Control is selected first; design is selected separately from structural value.
Random design	A random structural action is paired with EFE control.
Passive	No structural action.
Oracle	Ground-truth breach or routing mismatch selects the design action.

Table 1: Experimental conditions. The primary contrast is constitutive versus diagnostic only.

highly precise. This event does not change their roles. The drone must select a physical port swap. Successful transformation requires the realized routing to match the new precision-optimal partition and no collision. Both the old and new partitions can have zero factorization gap, so the predicted effect is on boundary precision and viability, not on Δ_{fac} .

5.2 Conditions

5.3 Frozen protocol and outcomes

The benchmark uses 20 held-out paired seeds for each of six conditions and three tasks, giving 360 missions. Exogenous random streams are paired by seed and task. The primary outcomes are task success and final architecture score

$$Q_{\text{arch}} = \exp[-\Delta_{\text{fac}}(\mathcal{C}_T)/8] \min\{\bar{\rho}_S(T), \bar{\rho}_A(T)\}. \quad (5.1)$$

Secondary outcomes are viability, collision, mean and final gap, role accuracy, blanket-set accuracy, exact final partition recovery, number of causal changes, and recovery delay. Viability is a bounded function of mean cross-track error, collision, and precision deficit.

A descriptive mediation analysis tests the proposed causal chain. With $X = 1$ for the constitutive condition, $M = Q_{\text{arch}}$, and Y equal to viability, task-adjusted ordinary least squares estimates

$$M = a_0 + aX + \Gamma T + \epsilon_M, \quad Y = b_0 + c'X + bM + \Lambda T + \epsilon_Y. \quad (5.2)$$

The indirect association ab is bootstrapped by paired seed. This analysis is supportive, not a substitute for the randomized primary contrast.

6 Results

6.1 Primary constitutive test

Figure 3 reports the frozen benchmark. The constitutive agent succeeded in $-$ % of construction trials versus $-$ % for diagnostic only; $-$ % versus $-$ % in stabilization; and $-$ % versus $-$ % in transformation. The paired final-gap difference was $-$ in construction and $-$ in stabilization. In transformation the final-gap contrast was zero by design, while the paired viability difference was $-$.

The primary comparison therefore answers the V4 objection directly. The diagnostic agent can update its model and can select nominal interventions, but it cannot make the desired conditional independencies true. The constitutive agent changes W_t or r_t and thereby changes the sampled path law.

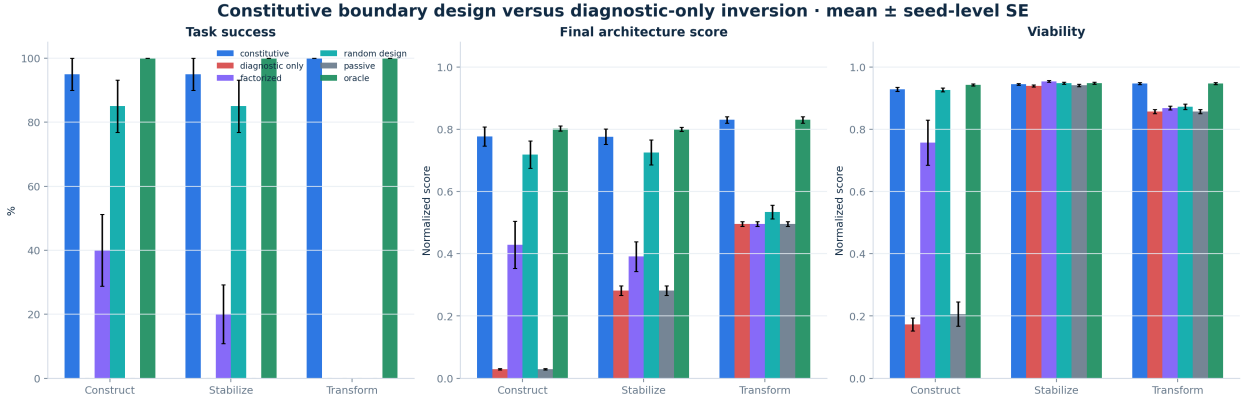


Figure 3: Task success, architecture score, and viability for the six conditions. Error bars are seed-level standard errors.

6.2 Construct, stabilize, transform

Figure 4 shows representative successful missions. Construction requires three discrete causal changes and ends at zero gap. Stabilization shows a seed-random breach followed by a selected repair. Transformation contains no forced role switch: the precision event changes which routing is useful, and a single swap performed by the drone changes the actual role map.

6.3 Complete role inference

V5 does not use four independent classifiers. The posterior ranges over complete partitions and produces marginals only by Equation (4.3). Figure 5 displays the four posterior fields in a transformation mission. Internal states are no longer an anchored target omitted from discovery; they are inferred jointly with sensory, active, and external roles.

6.4 Architecture and viability

Across the constitutive and diagnostic conditions, the task-adjusted indirect association through architecture score was $-$ (paired-seed bootstrap 95% interval $[-, -]$). This is consistent with the proposed chain

$$d_t \longrightarrow \mathcal{C}_{t+1} \longrightarrow \text{factorization/precision} \longrightarrow \text{viability}. \quad (6.1)$$

The result should be interpreted cautiously: the mediator is partly defined from simulator-level variables and the simple regression does not prove a general mechanistic law.

6.5 Ablations and process diagnostics

The ablation matrix in Figure 6 shows why success alone is insufficient. Random design can sometimes close one of a small number of breaches, but it does not solve targeted transformation; the factorized controller separates motor regulation from design and therefore misses interactions between structural and ordinary action. The oracle defines an attainable ceiling. The constitutive controller approaches that ceiling without access to the true role map.

Figure 7 makes the controller auditable. Risk favors stabilizing motor commands, ambiguity penalizes noisy predictions, epistemic value favors actions expected to distinguish role hypotheses, and structural value favors actions expected to close a bypass or improve routed precision. Because all terms are evaluated on the pair (u, d) , the value of a design operation depends on the control action with which it will be executed.

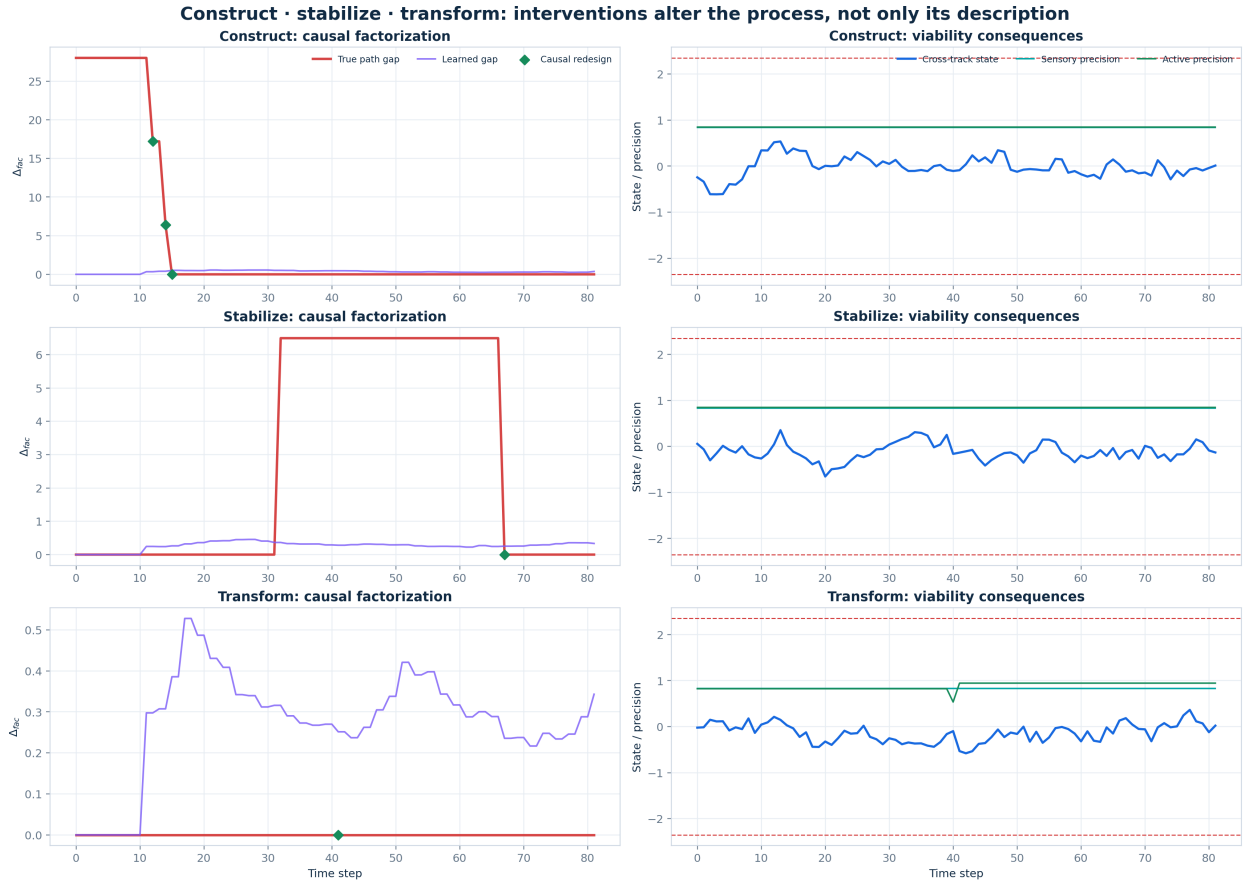


Figure 4: Representative missions for the three design verbs. Green diamonds mark actions that changed the causal matrix.

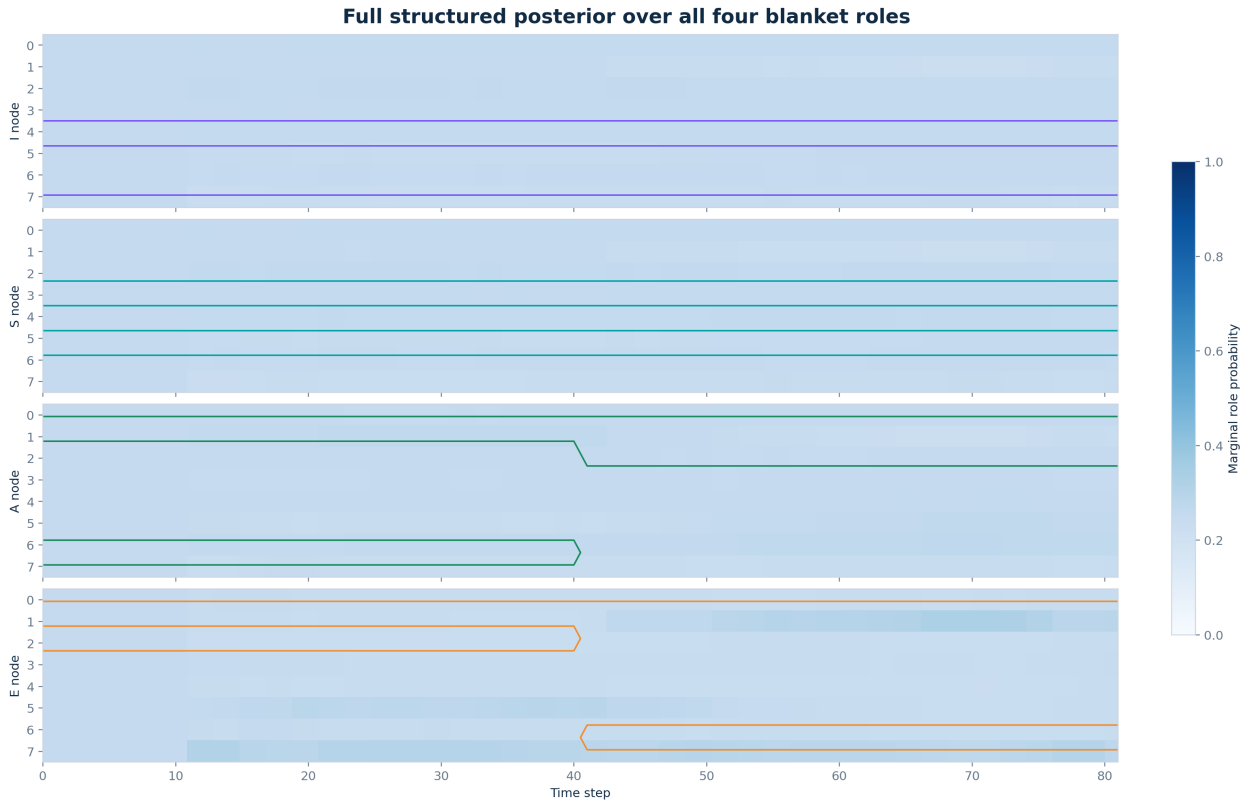


Figure 5: Posterior role probabilities for all eight modules and all four roles. Contours indicate the physical role assignment.

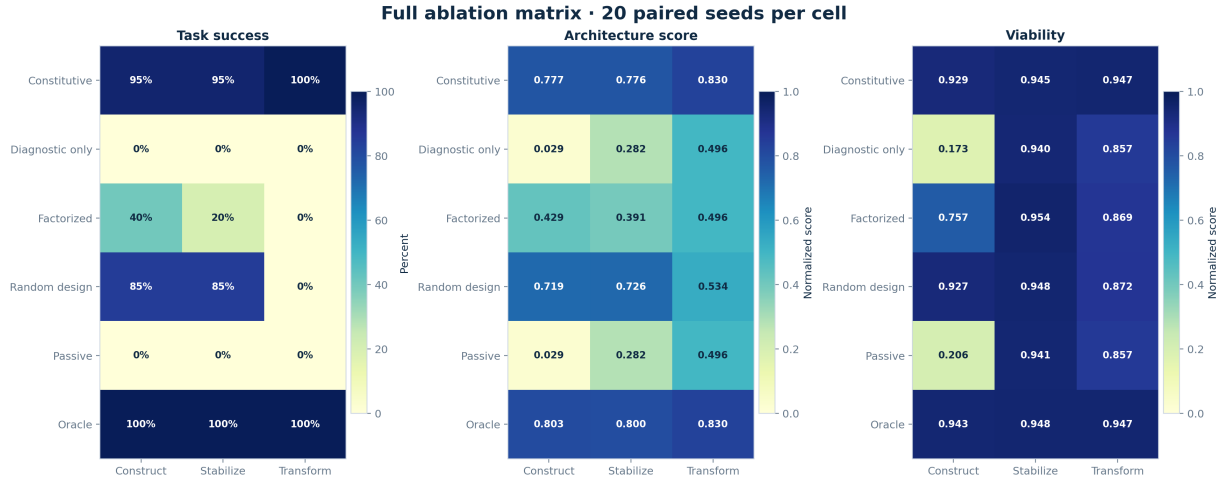


Figure 6: Part I ablation matrix. Each cell is a mean over the same 20 paired seeds. The architecture score distinguishes a merely surviving trajectory from one supported by the intended factorization and boundary precision.

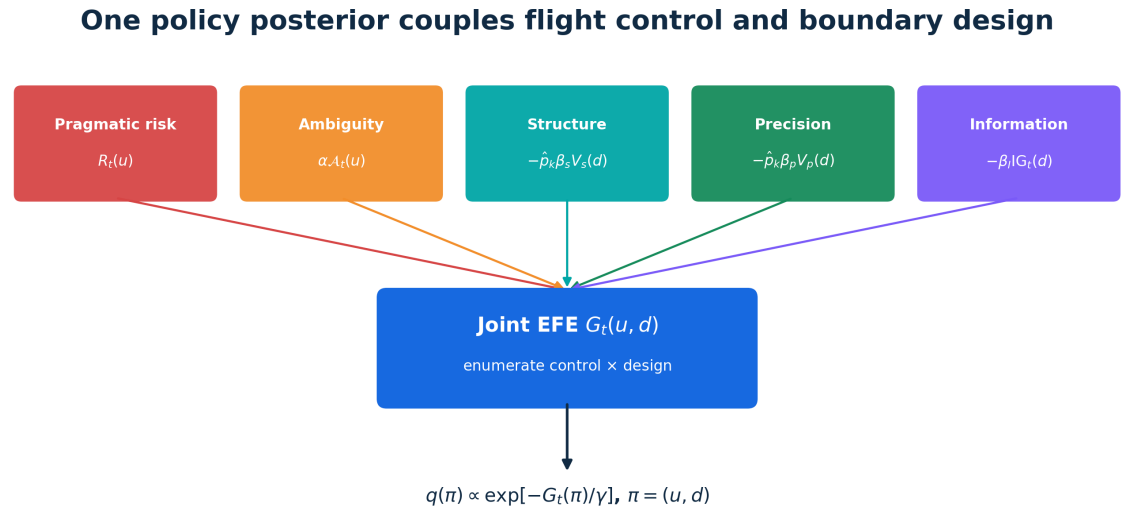


Figure 7: Implemented EFE decomposition and policy ranking in representative decisions. The chart reports the actual terms used by the controller rather than a post-hoc verbal classification.

The paired trajectories in Figure 8 hold the exogenous random stream fixed and vary only whether the selected structural action is allowed to modify $\mathcal{C}_t$. This comparison localizes the intervention: posterior updating occurs in both conditions, but only the constitutive arm changes the path-space gap and routed precision. Figure 9 adds posterior entropy, action timing, and intervention-model learning.

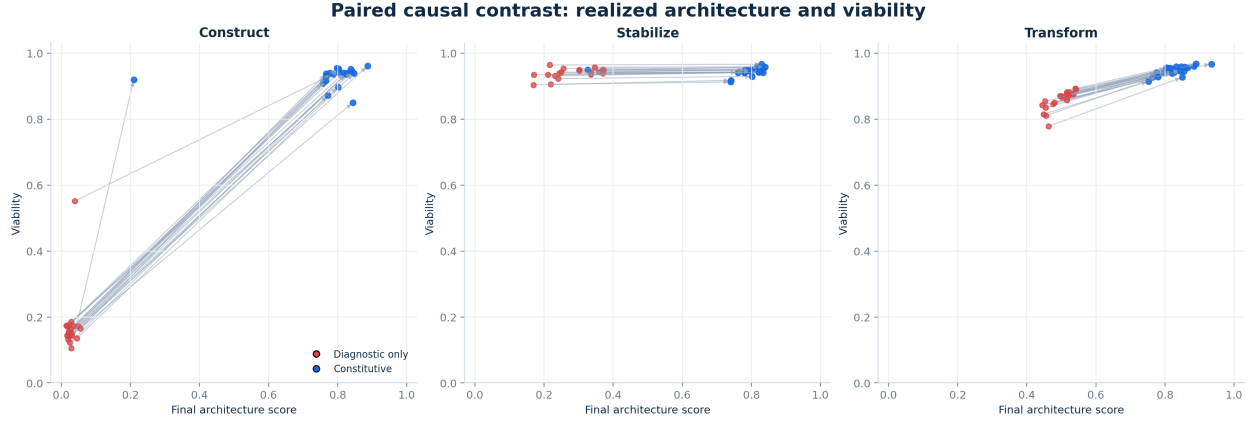


Figure 8: Paired causal-chain diagnostics. Within each seed, the constitutive and diagnostic runs share disturbances; the difference is the physical efficacy of design action.

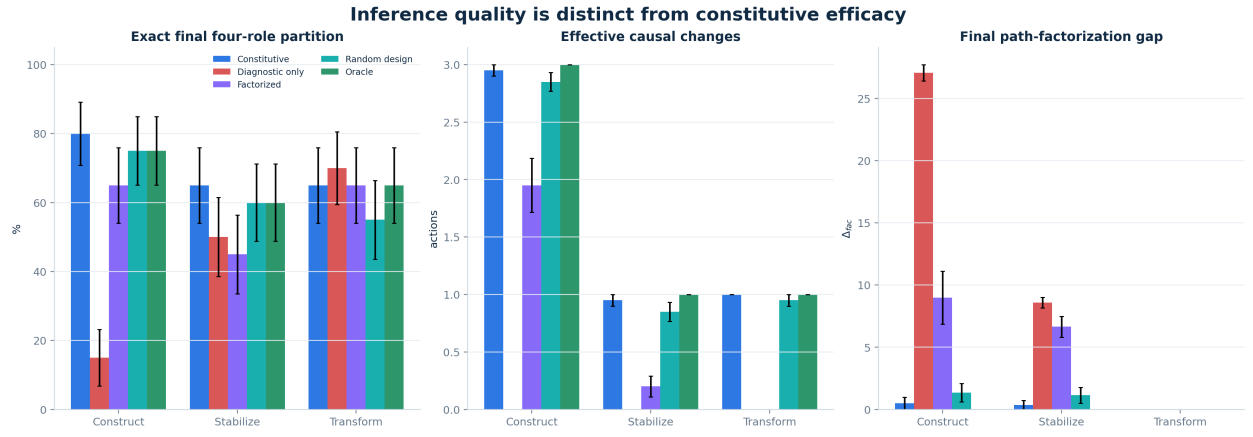


Figure 9: Inference and action diagnostics. Structural uncertainty need not vanish before useful action: policy value integrates posterior uncertainty, learned efficacy, and predicted architectural benefit.

Condition	Task	Success	Viability	Q_{arch}	Final Δ_{fac}	Role acc.	Exact
Constitutive	Construct	95.0%	0.929	0.777	0.498	65.2%	80.0%
Constitutive	Stabilize	95.0%	0.945	0.776	0.359	78.6%	65.0%
Constitutive	Transform	100.0%	0.947	0.830	0.000	71.6%	65.0%
Diagnostic only	Construct	0.0%	0.173	0.029	27.068	49.4%	15.0%
Diagnostic only	Stabilize	0.0%	0.940	0.282	8.589	76.3%	50.0%
Diagnostic only	Transform	0.0%	0.857	0.496	0.000	78.6%	70.0%
Factorized	Construct	40.0%	0.757	0.429	8.979	61.1%	65.0%
Factorized	Stabilize	20.0%	0.954	0.391	6.652	72.2%	45.0%
Factorized	Transform	0.0%	0.869	0.496	0.000	76.3%	65.0%
Random design	Construct	85.0%	0.927	0.719	1.349	66.2%	75.0%
Random design	Stabilize	85.0%	0.948	0.726	1.142	74.8%	60.0%
Random design	Transform	0.0%	0.872	0.534	0.000	72.3%	55.0%
Passive	Construct	0.0%	0.206	0.029	27.068	49.1%	25.0%
Passive	Stabilize	0.0%	0.941	0.282	8.589	74.3%	55.0%
Passive	Transform	0.0%	0.857	0.496	0.000	78.6%	70.0%
Oracle	Construct	100.0%	0.943	0.803	0.000	72.8%	75.0%
Oracle	Stabilize	100.0%	0.948	0.800	0.000	76.6%	60.0%
Oracle	Transform	100.0%	0.947	0.830	0.000	71.6%	65.0%

Table 2: Complete frozen benchmark. Values are means over 20 paired seeds.

Part II

Collective Boundary Design

7 From Coordination to Operational Individuation

Part II raises the scale of the question. Two drones may coordinate while remaining two independently bounded agents: they can exchange positions, follow a leader, or optimize a shared reward. None of these facts alone implies a collective Markov blanket. A stronger criterion is required. There must be a group-level partition whose factorization cannot be recovered from either drone’s evidence or unilateral action, and whose maintenance makes a measurable difference to joint performance.

The phrase “being one drone” is therefore used operationally and scale-relatively. It means that, for the modeled task and time horizon, some states distributed across two vehicles form a joint internal set; some form a joint sensory surface; some form a joint active surface; and the remaining states are external to the coupled unit. This is not a claim that the pair is an organism, a person, or an ontologically simple substance. It is a claim about the best-supported conditional organization of their trajectories and about their joint capacity to modify that organization.

Three tests distinguish a collective blanket from ordinary messaging:

1. **distributed evidence:** neither local observation window is sufficient for the complete role partition;
2. **joint efficacy:** no unilateral structural proposal can change the group organization;
3. **strategic consequence:** the enacted group organization changes payload, formation, energy, or viability outcomes relative to controls with the same communication bandwidth.

The experiment implements all three tests.

8 Collective Mathematical Model

8.1 Distributed states and candidate partitions

Let the two vehicles be $a \in \{A, B\}$. The candidate module set remains $V = \{1, \dots, 8\}$, with ownership map

$$h(j) = \begin{cases} A, & j \leq 4, \\ B, & j > 4. \end{cases} \quad (8.1)$$

Each owner initially contains one module of each role, but the role labels are permuted and never supplied to the learners. The collective hypothesis space is the same constrained set $\mathcal{M}$ in Equation (4.2); crucially, however, the evidence is distributed. Drone A observes modules $V_A = \{1, 2, 3, 4, 5, 6\}$ and drone B observes $V_B = \{3, 4, 5, 6, 7, 8\}$. Their overlap permits alignment, while the private modules prevent either local record from determining the complete partition.

Let D_t^a be the path fragment available to drone a . Local structural posteriors are

$$q_t^a(M) = \frac{p(D_t^a | M)p_0(M)}{\sum_{M' \in \mathcal{M}} p(D_t^a | M')p_0(M')}. \quad (8.2)$$

The agents exchange fixed-size sufficient-statistic messages m_t^a , not ground-truth roles or the physical adjacency matrix. A pooled transition estimate $\widehat{W}_t^{AB}$ is fitted from the union of communicated

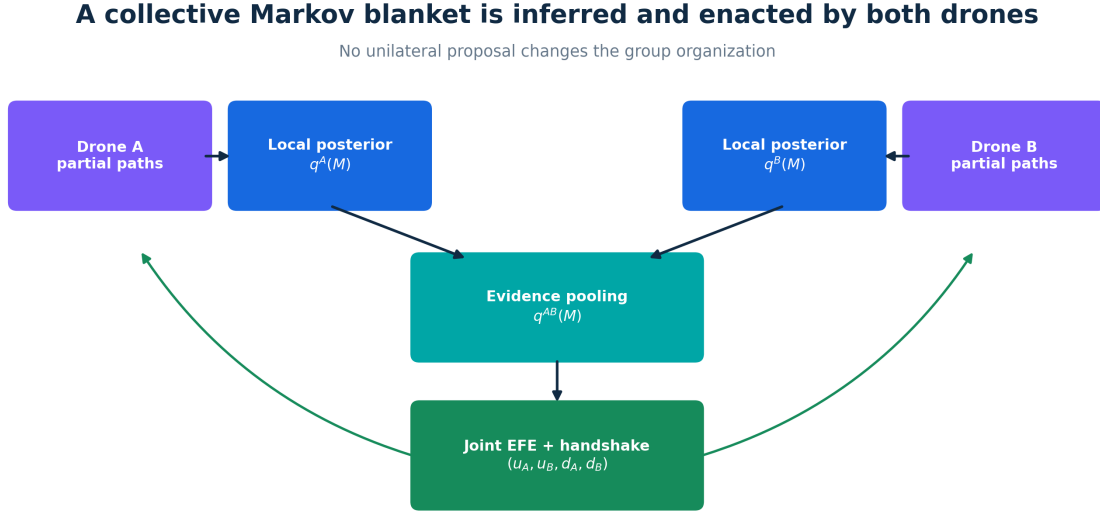


Figure 10: Collective inference and constitution. Each drone sees only partial paths and maintains a local structural posterior. Compressed evidence is pooled into a group posterior. A structural change occurs only when both agents select the same design proposal under joint EFE.

statistics. To avoid counting the prior twice, the group posterior is implemented as a logarithmic pool

$$q_t^{AB}(M) \propto \frac{q_t^A(M)q_t^B(M)}{p_0(M)} \exp\left[\lambda_g \ell(M; \widehat{W}_t^{AB})\right]. \quad (8.3)$$

All terms range over complete, permutation-aware role assignments. The group posterior is therefore not a vote over four independent labels.

8.2 A group-level causal organization

The collective causal state is

$$\mathcal{C}_t^{AB} = (r_t^{AB}, W_t^{AB}, L_t^\times, \rho_t, h), \quad (8.4)$$

where $L_t^\times$ contains unscreened links whose endpoints have different owners. A valid collective blanket screens group-internal paths from exterior paths conditional on the distributed boundary:

$$Y_{[0,T]} \perp\!\!\!\perp X_{[0,T]}^{AB} \text{ mid } B_{[0,T]}^{AB}, \quad B^{AB} = (S^{AB}, A^{AB}). \quad (8.5)$$

The collective factorization gap is

$$\Delta_{\text{fac}}^{AB}(\mathcal{C}_t) = \frac{1}{2\sigma^2} \left(\|W_{IE}^{AB}\|_F^2 + \|W_{EI}^{AB}\|_F^2 \right), \quad (8.6)$$

computed over the group partition rather than either owner partition.

This definition allows a multiscale fact that is central to the philosophical argument: a physical module owned by drone A can occupy an active role for the group, while a module owned by drone B can occupy a sensory role for the same group. Hardware ownership and blanket membership are different relations. The group boundary cuts across the two chassis.

8.3 Bilateral constitution

Each drone proposes a structural action $d_t^a \in \mathcal{D}^{AB}$. Cross-owner shielding removes a bypass between modules on different bodies; cross-owner routing exchanges two modules' group-level causal ports.

The collective transition is

$$\mathcal{C}_{t+1}^{AB} = \begin{cases} \mathcal{T}^{AB}(\mathcal{C}_t^{AB}, d_t^A, \omega_t), & d_t^A = d_t^B \neq d^0, \\ \mathcal{T}^{AB}(\mathcal{C}_t^{AB}, d^0, \omega_t), & \text{otherwise.} \end{cases} \quad (8.7)$$

Thus

$$\mathcal{T}^{AB}(\mathcal{C}, d, d^0, \omega) = \mathcal{T}^{AB}(\mathcal{C}, d^0, d, \omega) = \mathcal{T}^{AB}(\mathcal{C}, d^0, d^0, \omega), \quad (8.8)$$

while two matching non-null proposals can change W^{AB} or r^{AB} . This handshake is not meant as a universal model of social agreement. It is an experimental identifiability device: it makes collective causal efficacy irreducible to one privileged agent.

8.4 Shared payload and formation dynamics

Let y_t^A and y_t^B be lateral positions. The task variables are shared payload displacement and formation error,

$$p_t = \frac{1}{2}(y_t^A + y_t^B), \quad f_t = (y_t^B - y_t^A) - f^*, \quad (8.9)$$

where f^* is the preferred separation. The two control inputs affect the center and relative modes:

$$p_{t+1} = \alpha_p p_t + \delta_p(\bar{u}_t + w_t) + \xi_t^p, \quad (8.10)$$

$$f_{t+1} = \alpha_f f_t + \delta_f(u_t^B - u_t^A) + \xi_t^f, \quad (8.11)$$

with $\bar{u}_t = (u_t^A + u_t^B)/2$. Sensory precision changes uncertainty in both estimates; active precision changes effective actuation. Cross-owner breaches add correlated disturbances and energetic compensation. The group viability score is a bounded function of payload RMSE, formation RMSE, collision, energy, factorization gap, and routed boundary precision.

8.5 Joint expected free energy and agreement

A collective policy is $\pi^{AB} = (u^A, u^B, d^A, d^B)$. The implemented group EFE is

$$G_t^{AB}(\pi^{AB}) = R_t^{\text{payload}} + R_t^{\text{formation}} + R_t^{\text{energy}} + \alpha \mathcal{A}_t^{AB} \quad (8.12)$$

$$+ c(d^A) + c(d^B) - \beta_M \mathbb{E}[\Delta Q_{\text{arch}}^{AB}] - \beta_I \mathbb{I}(M; o_{t+1}^{AB} | \pi^{AB}) + \kappa \mathbf{1}[d^A \neq d^B]. \quad (8.13)$$

The final term prices disagreement because unmatched proposals consume a design opportunity but cannot change the organization. Each drone evaluates candidate pairs using q_t^{AB} and the learned family-level efficacy model, then samples from a shared softmax

$$q(\pi^{AB}) \propto \exp[-G_t^{AB}(\pi^{AB})/\gamma_g]. \quad (8.14)$$

Agreement is therefore endogenous to policy selection rather than imposed after two independent decisions.

9 Collective Experimental Design

9.1 Tasks

Assembly. The pair begins with cross-owner internal-external bypass links. It must infer which cross-body couplings violate the group partition and agree on the corresponding shields. This constructs a group boundary that was absent at mission start.

Condition	Operational definition
Collective constitutive	Pooled posterior, joint EFE, bilateral handshake, and physically effective group-level design.
Collective diagnostic	Same inference, messages, and nominal agreement, but design cannot alter $\mathcal{C}^{AB}$.
Individual constitutive	Each drone acts from its local posterior; matching proposals can act, but there is no pooled group model.
Fixed team	Shared controller with no structural action and a fixed engineered interface.
Communication only	Same aggregate message budget, used for raw state sharing without a role posterior or boundary model.
Random joint	Bilaterally matched structural actions sampled without EFE.
Oracle collective	Ground-truth breach or precision mismatch selects the bilateral action; this is an upper bound, not a realizable learner.

Table 3: Part II conditions. Communication-only is the critical control for the possibility that bandwidth, rather than blanket inference, produces the strategic advantage.

Repair. The pair begins with a clean group boundary. At a random time a cross-owner bypass opens. A nonspecific integrity alarm enables structural action but does not identify the link or reveal role labels. The drones must localize and close the breach before mission end.

Reconfiguration. A currently useful boundary module loses precision while a precise external module owned by the other drone becomes available. The factorization gap remains zero, so generic shielding cannot solve the task. The agents must agree on the appropriate cross-owner port exchange. This is the strongest discrimination between purposive design and random structural action.

9.2 Conditions and controls

The frozen protocol uses 20 held-out paired seeds for each of seven conditions and three tasks: 420 missions of 90 steps. The seed offset (2000), software versions, and full mission-level observations are stored with the results. Primary outcomes are task success and group architecture score. Secondary outcomes are group viability, payload and formation RMSE, energy, final factorization gap, role and blanket-set accuracy, exact partition recovery, agreements, and realized causal changes.

10 Collective Results

10.1 Constitution and strategic advantage

The collective constitutive agent succeeds in 80% of assembly missions, 90% of repair missions, and 100% of reconfiguration missions (Figure 11). Both the collective diagnostic and equal-bandwidth communication controls score 0% on all three tasks. Against communication only, the paired success advantages are 0.80 ± 0.092 , 0.90 ± 0.069 , and 1.00 ± 0.000 ; group viability advantages are 0.114 ± 0.018 , 0.052 ± 0.004 , and 0.111 ± 0.004 .

Compared with individual constitutive agents, the collective success advantages are 0.55 ± 0.114 in assembly, 0.75 ± 0.099 in repair, and 0.60 ± 0.112 in reconfiguration. The corresponding architecture-score advantages are 0.373 ± 0.071 , 0.414 ± 0.056 , and 0.224 ± 0.043 . These comparisons answer RQ2 narrowly: pooled structural evidence plus bilateral design outperforms local blankets under the simulated partial-observability regime.

10.2 Why messaging alone is not enough

The communication-only agents receive the same aggregate bandwidth and often obtain competitive payload or formation estimates. Nevertheless they cannot close a bypass or reroute a boundary port because their messages are not organized as evidence about a candidate partition and their controller has no causal boundary target. Figure 12 shows the paired viability comparison. Almost every seed lies above the identity relation in favor of collective boundary design.

The distinction is not that communication is unnecessary. Evidence pooling is itself communicative. The point is that bandwidth is organized by a model of causal roles and connected to actions that can change those roles. Collective individuation is a use of communication, not an alternative to it.

10.3 Random constitution and directed transformation

Random joint action succeeds in 60% of assembly and 75% of repair missions. This is expected in a small action space: repeated matched shielding can eventually hit a breach. Random action nevertheless succeeds in 0% of reconfiguration missions, where the target is one precision-improving cross-owner swap among multiple causally valid partitions. Thus the data do not show that EFE is necessary for every causal modification. They show that EFE supplies selectivity when several structural changes are possible and only one realizes the preferred interface.

10.4 Time-resolved examples

Figure 14 displays one successful mission per task. In assembly the factorization gap collapses after agreed shielding. In repair the gap rises at the exogenous breach and returns to zero only after a bilateral action. In reconfiguration the gap remains zero throughout, but the agreed swap improves the routed precision needed for payload and formation control. These different signatures are why construct, stabilize, and transform cannot be collapsed into a single “adaptation” score.

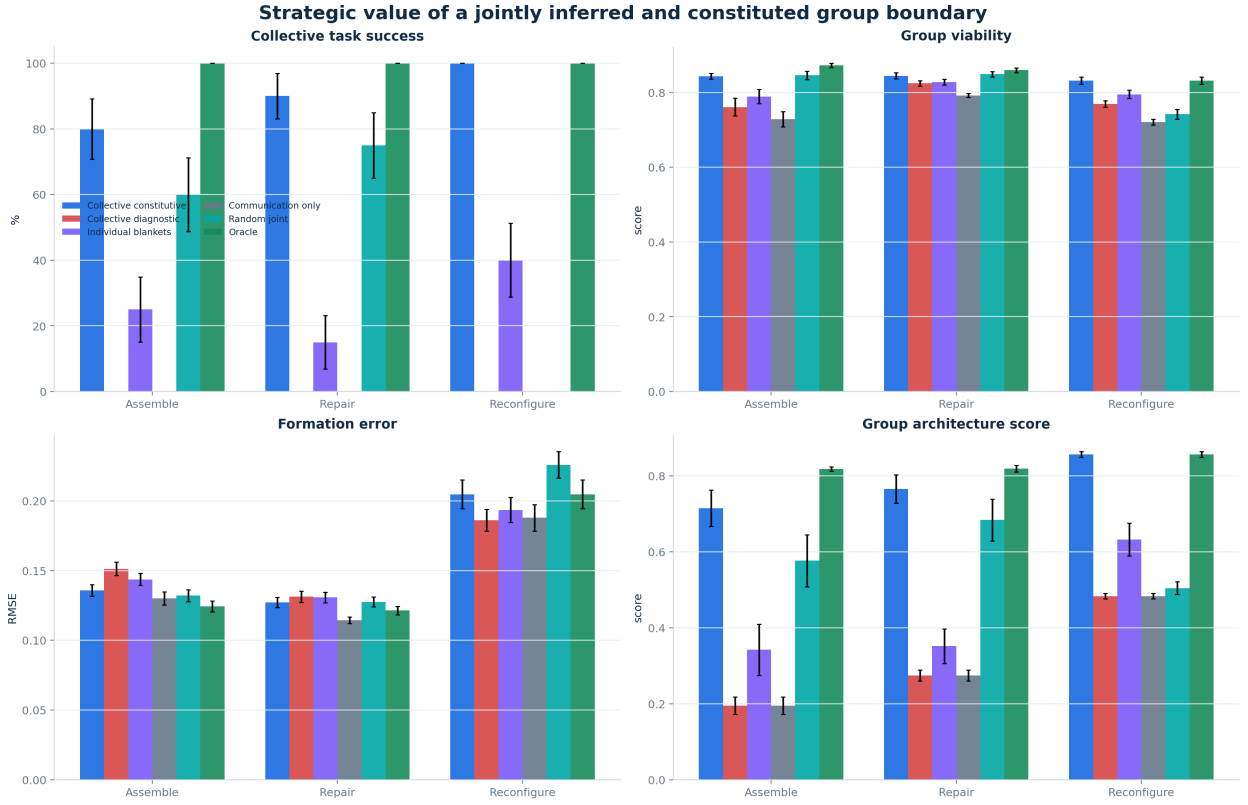


Figure 11: Collective benchmark. Error bars are seed-level standard errors. Success requires the targeted group organization, not merely collision-free motion. The oracle marks the attainable ceiling.

Condition	Task	Success	Viability	Q_{arch}^{AB}	Payload RMSE	Formation RMSE	Energy
Collective constitutive	Assemble	80%	.843	.715	.183	.136	8.69
	Repair	90%	.844	.766	.183	.127	8.62
	Reconfigure	100%	.832	.857	.189	.205	8.69
Collective diagnostic	Assemble	0%	.761	.195	.276	.151	12.84
	Repair	0%	.825	.274	.198	.131	10.08
	Reconfigure	0%	.770	.483	.180	.186	8.17
Individual constitutive	Assemble	25%	.789	.342	.244	.144	11.44
	Repair	15%	.828	.352	.197	.131	9.81
	Reconfigure	40%	.795	.633	.183	.194	8.38
Communication only	Assemble	0%	.729	.195	.200	.130	19.29
	Repair	0%	.792	.274	.107	.114	15.93
	Reconfigure	0%	.721	.483	.118	.188	15.17
Random joint	Assemble	60%	.846	.577	.177	.132	8.61
	Repair	75%	.849	.684	.179	.128	8.32
	Reconfigure	0%	.742	.504	.208	.226	10.03
Oracle collective	Assemble	100%	.873	.818	.158	.124	6.64
	Repair	100%	.860	.819	.172	.121	7.52
	Reconfigure	100%	.832	.857	.189	.205	8.69

Table 4: Selected complete Part II outcomes. Means are over 20 paired seeds. The fixed-team condition is numerically identical to collective diagnostic for the reported variables and is retained in the archived CSV.

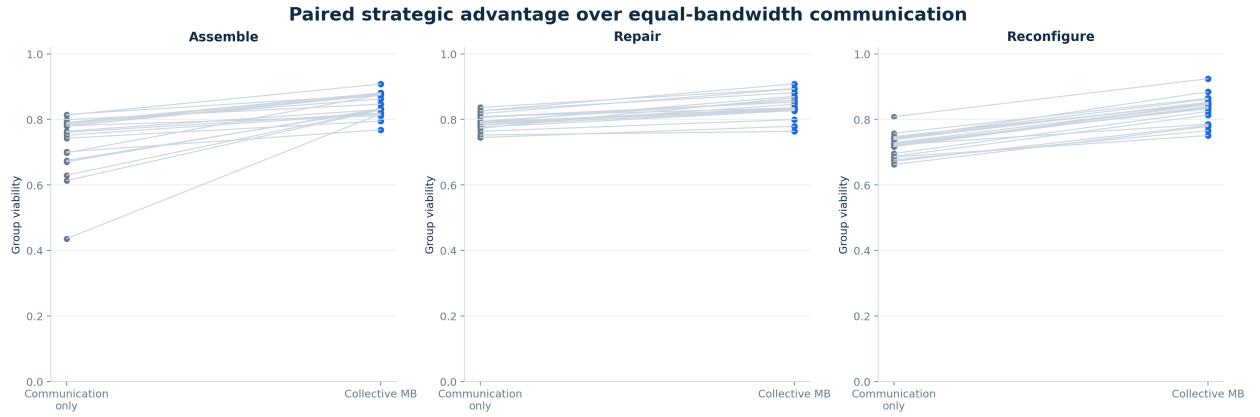


Figure 12: Paired strategic advantage over equal-bandwidth communication. Lines connect the same exogenous seed. The comparison isolates the value of representing and constituting a group boundary from the value of sharing data as such.

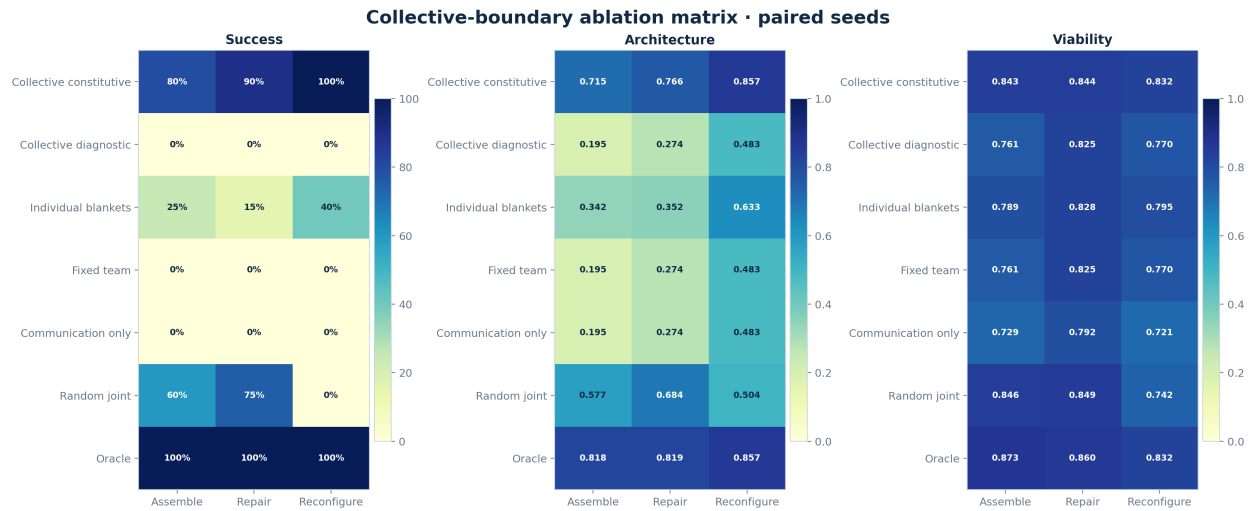


Figure 13: Collective ablation matrix. The reconfiguration column is the discriminating test for purposive boundary design: factorization alone does not choose among multiple zero-gap boundaries.



Figure 14: Representative collective missions. Green diamonds mark bilateral changes to the group organization. Red traces are group factorization gaps; violet traces are posterior entropies; right panels show shared payload and formation trajectories.

Part III

Synthesis

11 Discussion

11.1 What the model now demonstrates

Part I establishes more than detection or tracking in its simulated domain. First, a drone-selected action changes the physical causal organization under an otherwise fixed exogenous realization. Second, that change can remove a path-law coupling or reassign a physical module to a different interface role. Third, the change improves a preregistered architecture score and, in the construction and transformation settings, viability. Fourth, the effect disappears in the diagnostic-only control.

These points operationalize Possati's inversion. The agent does not merely discover where a boundary already is. It uses the discovered organization to select material operations that make another boundary true. Construction, stabilization, and transformation are separated because they are different claims: closing initial bypasses, repairing a later breach, and changing a valid boundary because another valid boundary better supports the task.

Part II adds a distinct result. The successful unit of structural action is not either vehicle by itself. Local learners have incomplete evidence; unilateral proposals are causally inert; and the target partition crosses hardware ownership. The enacted organization improves group viability over an equal-bandwidth controller and improves targeted success over two local constitutive agents. In this bounded sense, the pair becomes a single operational agent for the shared task: it acquires a joint inferential interior and a jointly controlled interface.

11.2 Why the book connection is substantive

The model clarifies the philosophical claim in three ways. First, a Markov blanket is treated as a structuring of possible state paths, not as an object surface. Second, inversion is doubled: Bayesian inversion estimates the current structure, while design inversion acts on the world so that a preferred structure becomes realized. Third, precision, curiosity, and prediction cease to be exclusively mental or computational notions. They become properties of material selection, intervention, and routing.

This is the precise point at which Possati differs from accounts that use active inference to optimize an artifact inside a fixed system boundary. Niche construction changes environmental states; interaction design changes the observations available to a user; explainable active inference changes access to internal beliefs. Possati's strong proposal concerns the causal partition that makes "internal," "sensory," "active," and "external" roles possible in the first place. V5 is a deliberately minimal realization of that claim.

The collective extension sharpens the book's thesis. If boundaries were fixed surfaces, two drones could only cooperate across already constituted interfaces. If blankets are organizations of paths, however, cooperation can itself have a boundary that is discovered and materially sustained. The interface is then not simply between drone *A* and drone *B*; it can run through both of them, separating group-internal dynamics from the shared environment. Possati's inversion becomes multiscale: design does not only alter what an existing individual senses and does, but can alter which individual is the relevant unit of sensing and doing.

11.3 Does the pair literally become one drone?

The experiment warrants an operational, not an unrestricted ontological, answer. It supplies four pieces of evidence for group-level individuation: a nontrivial collective partition, distributed epistemic access, irreducibly bilateral causal control, and a strategic consequence of maintaining the partition. These criteria exclude mere co-location and simple data exchange. They do not prove phenomenal unity, biological autonomy, indefinite self-maintenance, or persistence across arbitrary tasks.

The most defensible formulation is therefore: the two vehicles instantiate one *task-relative control unit* over the mission horizon. Whether that unit should count as one artifact without qualification depends on further criteria—material integration, counterfactual robustness, persistence, normative closure, and the capacity to recreate its own enabling components. The present Markov-blanket evidence is neither trivial nor metaphysically decisive.

11.4 Alternative explanations and the role of the controls

Five alternatives are addressed experimentally. Better results are not due only to model updating, because the diagnostic control updates without constituting. They are not due only to communication, because bandwidth is matched. They are not due only to structural actuation, because random joint action fails targeted transformation. They are not reducible to two independent local blankets, because the individual condition lacks the full evidence and performs worse. Finally, they are not due to an exogenous role switch: disturbances alter couplings or precision, while the role exchange is selected by the agents.

Not every alternative is excluded. The simulator and success criteria share the same formal vocabulary; the controller is therefore evaluated in a world whose relevant regularities it can represent. The action set is small, the message topology is engineered, and both agents use the same model class. Robustness to model misspecification and heterogeneous learners remains open.

11.5 What remains unproven

Six limitations delimit the result. First, the eight candidate modules are declared. The system does not discover an unbounded ontology. Second, the action repertoire is engineered: shielding and port exchange are known affordance types even though their success is learned. Third, the linear-Gaussian gap in Equation (4.7) is exact only for the simulator’s chosen local path-law comparison. Nonlinear, partially observed hardware requires richer estimators. Fourth, the EFE score uses engineering weights and a short horizon. It demonstrates the conjunction of pragmatic, ambiguity, and epistemic terms, not a unique first-principles controller. Fifth, the collective handshake is a designed institution rather than an emergent communication protocol. Sixth, all results come from a linear-Gaussian simulator and 20 seeds per cell; they establish internal proof-of-concept validity, not ecological generality.

The learned intervention model is also intentionally small. A stronger V6 would learn action-conditioned changes to the full transition matrix, represent uncertainty over hardware affordances, and evaluate counterfactual path laws over longer horizons. A hardware experiment would require safe reconfigurable sensor buses, redundant actuators, online causal estimation, and a conservative supervisory layer. A stronger collective version would also learn who participates in the group, the communication graph, and the conditions under which the group should dissolve.

12 Conclusion

The decisive question was whether “boundary design” named a causal operation or only a re-description. V5 makes that question testable by separating the inferred blanket from the physical organization, allowing design actions to modify the latter, and comparing them against identical diagnostic actions that cannot. Within the declared simulator, the result is positive across construction, stabilization, and transformation. The second experiment shows that the same logic can cross physical bodies: two drones can pool evidence, agree on a structural operation, and enact a task-relative group boundary that improves strategic performance. The proper claim is narrow but no longer metaphorical. A single drone can redesign its statistical interface, and a pair can constitute an operational interface that neither can maintain alone.

Reproducibility Statement

The repository contains the complete source, unit tests, fixed seed offsets, mission-level CSVs, paired effects, mediation output, run metadata, and figure generation code. Part I is reproduced with `python run_experiment.py -seeds 20 -seed-offset 1000 -workers 6`; Part II uses `python run_collective_experiment.py -seeds 20 -seed-offset 2000 -workers 6`. The diagnostic-only contrasts, equal-bandwidth communication control, paired seeds, and all three tasks must be retained in replication.

Technical Appendices

A Path-Space Factorization and the Linear Gap

This appendix states the limited derivation behind Equation (4.7). Consider two path coordinates X and Y after conditioning on a proposed blanket path B . Let P denote a diffusion with direct residual cross-drift $g_t = (W_{XY}Y_t, W_{YX}X_t)$ and let P^0 denote the same diffusion with that cross-drift removed. If both laws share nonsingular diffusion matrix σI and satisfy the usual absolute continuity and integrability conditions, Girsanov’s theorem gives

$$D_{\text{KL}}(P\|P^0) = \frac{1}{2\sigma^2} \mathbb{E}_P \int_0^T \|g_t\|_2^2 dt. \quad (\text{A.1})$$

For standardized coordinates with unit second moment and a locally constant drift matrix,

$$\frac{1}{T} D_{\text{KL}}(P\|P^0) \approx \frac{1}{2\sigma^2} \mathbb{E}[\|W_{XY}Y\|_2^2 + \|W_{YX}X\|_2^2] \quad (\text{A.2})$$

$$\approx \frac{1}{2\sigma^2} (\|W_{XY}\|_F^2 + \|W_{YX}\|_F^2). \quad (\text{A.3})$$

The right-hand side is the implemented Δ_{fac} . It vanishes when the proposed blanket screens direct cross-drift in the simulator. It is a local relative-entropy-rate proxy, not an estimator of arbitrary nonlinear conditional mutual information. Autocorrelation, latent common causes, or state-dependent diffusion would require a richer path-law model.

For the stabilization task, the paired effect is identifiable because the same disturbance realization is used in both arms. Let $\omega_{0:T}^{(s)}$ be seed s ’s exogenous stream. Then

$$\delta_s = \Delta_{\text{fac}}(\mathcal{C}_T^{\text{constitutive}}(\omega^{(s)})) - \Delta_{\text{fac}}(\mathcal{C}_T^{\text{diagnostic}}(\omega^{(s)})) \quad (\text{A.4})$$

isolates whether the selected design operation is permitted to change the causal organization. The reported standard error is $\text{sd}(\delta_s)/\sqrt{20}$.

B Exact Partition Posterior and Permutation Awareness

The eight-module role space is generated by nested combinations. Choose two modules for I, two from the remainder for S, two for A, and assign the final two to E. This yields

$$\binom{8}{2} \binom{6}{2} \binom{4}{2} \binom{2}{2} = 2520 \quad (\text{B.1})$$

hypotheses. Because the members within a role are exchangeable, evaluation compares role sets rather than arbitrary node order. Exact partition accuracy is

$$\mathbf{1} \left[\arg \max_{M \in \mathcal{M}} q_T(M) = M_T^{\text{true}} \right], \quad (\text{B.2})$$

and blanket-set accuracy averages classification of membership in $B = S \cup A$. Exact recovery is reported separately from task success because an agent can select a useful intervention under posterior uncertainty without recovering every role label.

The structural template score uses magnitude to detect adjacency and learned control/noise terms to orient the cycle. This prevents a purely symmetric graph match from arbitrarily exchanging sensory and active labels. A hazard mixture

$$\tilde{q}_t(M) = (1 - h)q_t(M) + h/|\mathcal{M}| \quad (\text{B.3})$$

keeps alternatives alive after structural change. In the collective model, the same hypothesis index is shared between local and pooled posteriors, so logarithmic pooling cannot combine incompatible label spaces.

C Algorithms

C.1 Individual constitutive inversion

1. Initialize a uniform posterior over $\mathcal{M}$, a rolling transition buffer, and Beta(1, 1) efficacy beliefs for shielding and swapping.
2. Observe the current module vector, physical control variables, and precision diagnostics; append the transition to the buffer.
3. Fit $\widehat{W}_t$, control response, and residual covariance by ridge regression; score all 2520 partitions and update $q_t(M)$.
4. Enumerate admissible motor–design pairs (u, d) . For each pair, predict pragmatic risk, ambiguity, structural value, precision value, and information gain under $q_t(M)$ and the efficacy posterior.
5. Sample or select a pair from $q(\pi) \propto e^{-G(\pi)/\gamma}$. Apply u . If design is enabled, apply the persistent shield or port swap.
6. Compare the pre/post causal estimate, update the efficacy posterior, and repeat until the mission terminates.

C.2 Collective constitutive inversion

1. Each drone updates $q_t^a(M)$ from its partial path window and encodes a fixed-size sufficient-statistic message m_t^a .
2. Pool the messages, fit $\widehat{W}_t^{AB}$, and form $q_t^{AB}(M)$ by Equation (8.3).

3. Enumerate joint motor commands and admissible cross-owner design actions. Score payload, formation, energy, ambiguity, architectural value, information gain, and disagreement.
4. Each drone receives the same group-policy distribution but issues its own proposal. Apply a structural transition only if the two proposals match.
5. Broadcast whether a causal change occurred, update the shared efficacy belief, and continue local and pooled inference.

This architecture is centralized only at the level of statistic pooling and joint policy evaluation. Observations and proposals remain owner-indexed. A fully decentralized successor would approximate the logarithmic pool by consensus and negotiate policy without a shared evaluation process.

D Parameter Register

Component	Value	Function
Modules / roles	8 / 4	Two exchangeable modules per role.
Partition hypotheses	2520	Exact constrained posterior.
Part I horizon	96 steps	Individual construct, stabilize, transform.
Part II horizon	90 steps	Collective assemble, repair, reconfigure.
Part I seeds	20, offset 1000	Paired frozen benchmark; 360 missions.
Part II seeds	20, offset 2000	Paired frozen benchmark; 420 missions.
Spectral-radius cap	0.82	Keeps module dynamics stable.
Collision threshold	$ y > 2.35$	Individual safety failure.
Efficacy prior	Beta(1, 1)	Uninformative family-level prior.
Design budget	10	Upper bound on collective attempts.
Local views	6 of 8 modules	Overlapping, individually incomplete evidence.
Ownership	4 + 4 modules	One initial local instance of each role.
Handshake	exact match	No unilateral structural efficacy.

Table 5: Parameters most relevant to interpretation. All numerical controller weights and simulator constants are defined in the released source rather than hidden in the manuscript.

E Outcome Definitions and Statistical Discipline

Individual construction succeeds if the final gap is below 10% of its initial value and no collision occurs. Stabilization succeeds if the final gap is below 0.25 and no collision occurs. Transformation succeeds if the realized partition equals the post-event precision-optimal routing and no collision occurs. Collective tasks use the analogous group-level criteria and require a realized bilateral causal change.

No null-hypothesis significance tests are used to turn the simulation into a population claim. The paper reports paired mean differences and standard errors to show seed-level stability. The 20 seeds are computational replications under a fixed simulator distribution, not samples from a real drone population. Hyperparameters were developed on separate quick-run seed ranges; the reported offsets were run once as the frozen evaluation set.

The mediation calculation is explicitly descriptive. Architecture score is a designed mediator and includes simulator-level quantities unavailable on hardware. Its bootstrap interval tests stability of the proposed chain inside this model; it does not identify a universal causal mediation effect.

F Reproducibility and Test Register

The release contains machine-readable mission-level data, summaries, paired effects, metadata, every plotted figure, and source code for both experiments. Ten unit tests verify partition enumeration and normalization, constitutive versus diagnostic transitions, learned efficacy updates, joint policy selection, collective posterior normalization, and the two invariants most important to Part II: a unilateral action cannot change the group boundary, whereas a matching bilateral action can.

The recommended verification sequence is:

1. install the pinned Python requirements;
2. run `python -m unittest discover -s tests -v`;
3. reproduce Part I and Part II with the commands in the reproducibility statement;
4. compare generated metadata, summary CSVs, and figure hashes with the archived outputs;
5. compile the manuscript with three passes of `pdflatex`.

The code deliberately exposes the distinction between epistemic and physical state. Posterior objects are stored in the inference classes; causal role maps, drift matrices, precision vectors, and unscreened links live in the world classes. A regression that accidentally lets the learner write the ground-truth role map would therefore cross an explicit software boundary and fail the constitutive-control tests.

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